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MTH 481
Course Project 2

Linear Programming Project:
Floor Space Optimization for a Multi-Tier Pinball Venue

Background

My name is Matthew Petrarca, I am currently completing my undergraduate degree in Computer Science at the University of Rhode Island, and am formerly the #1 ranked IFPA (International Flipper Pinball Association) competitive pinball player in the state of Rhode Island. Prior to returning to college, I was previously employed as a pinball technician by a bar in Westerly named “Flipside”, which unfortunately closed during the pandemic. Since then I have been making considerations towards potentially opening my own pinball bar in the future, however, with an alternative operations model in mind.

Originally, I had taken a hard look at the limitations that a pinball bar, such as Flipside, would face and wondered if there was potential for a more viable business model that revolved around a general space containing an assortment of pinball machines. For example, given the success of entertainment business models in recent years, such as the bar-cades, TopGolf, and various axe-throwing venues that have emerged, I wondered if there was potential for pinball to take on a newer presentation style or format. For example, adding private rentable pinball bays or rooms for parties and groups, similar to TopGolf, bowling alleys, or axe throwing lanes.

Introduction

Given my personal experience over two years of casual to competitive pinball in the State of Rhode Island, I began to detail what I would consider to be an ideal, pinball-focused venue. Without getting into too much detail over what is now a 20+ page business plan, I decided on creating a premium, destination pinball venue structured around four distinct zones: a Freeplay floor accessible with a flat entry fee, a Premium play room that offers rare, limited edition, or top-condition machines at a $1 per game, private rental bays for groups/events, and a general bar area. Then, the focus shifted to which of these sources of revenue required the highest priority and resources, in terms of supporting the business model.

For example, if the bar sales generate the most significant amount of revenue, then, from a business perspective, space and resources must be allocated to maximize the bar zone. If, on the other hand, any of the other zones, such as the private bays, entry fees, or $1 per game sections, should generate the majority of the revenue then the focus becomes maximizing those revenue options instead. This led to the creation of an HTML-based floor space simulator that could scale based on allocated square footage across each of the different zones. This allowed for the space to be given a baseline of machines in the FreePlay, Premium, and Private Bay sections, while also displaying real-time annual revenue projections. After some tinkering

and initial exploratory analysis, several viable floor layouts were derived as possible implementations, but there was no way of definitively knowing which was the best option.

So, ultimately, the question became, how can I optimize the floor space of a pinball venue, such that the multi-zoned business model generates the most revenue overall? Well, as it turned out, I was enrolled in a course that would teach me to do just that. So, for my project I created a Linear Programming model to determine the optimal square footage for each revenue-generating zone, given the physical constraints of the space and the minimum viable size requirements for each zone to function as intended.

Revenue Model

The venue is modeled on a fixed 3,500 square foot location with a fixed back-of-house allocation for storage, maintenance, and general operations. Before formulating the Linear Programming model, annual revenue coefficients must be derived for each zone. These coefficients represent the revenue generated per square foot per year and serve as the objective function weights. The derivation draws on modeled estimates that are grounded in industry benchmarks, competitive venue pricing, and direct observational data from local venues featuring pinball machines in the greater Providence, Rhode Island area. The figures are conservative estimates intended to serve solely as a viable, but not exceptional first-year operation.

Fixed Parameters

- The following parameters are treated as constraints throughout the model:

Parameter	Value	Description
S	3,500 sq ft	Total venue square footage
V	40 visitors/day	Average daily visitors
d	364 days/year	Annual operating days (generous)
e	$12.50	Average entry fee per visitor
p	$8.00	Blended premium play spend per visitor
b	$12.00	Average bar spend per visitor
u	0.50	Bay utilization rate
r	$95.00	Average bay booking revenue
w	3.5 bookings/week	Maximum bookings per bay per week at full utilization
f	500 sq ft	Fixed: bar + entry + bathrooms (minimum functional allocations)

h	320 sq ft	Fixed: back of house (machine storage, repairs, operations/utility)
c	0.12	Circulation factor applied to total allocated space

Space Allocation Standards:

Each pinball machine is allocated 65 square feet of floor space. This figure reflects the premium spacing standard of the venue: a standard arcade allocates 50 sq ft per machine, but the premium model requires additional clearance for comfortable play, nudging, spectating, and aisle traffic; all necessary for a premium playing experience. Each private bay occupies 220 sq ft, sized for a machine, bar-height table, and seating for up to four.

The baseline square footage machine counts of 15 FreePlay machines, 6 Premium machines, and 3 private bays were determined through prior trial exploration, using the previously mentioned interactive HTML floor space simulator that was built during the business planning phase. The simulator allowed for real-time adjustment of machine counts, bay count, visitor assumptions, and total square footage with live revenue output.

Testing configurations across the feasible range revealed three key observations that directly informed the LP formulation: entry fee revenue scaled with visitor counts rather than machine count (thus adding additional machines to the FreePlay area produced no revenue gain), private bay revenue demonstrated a structural ceiling regardless of bay count or utilization rate, and bar revenue consistently dominated all other streams at higher attendance levels, scaling directly with visitor count. These observations established the upper bounds used in the LP constraints and confirmed that the coefficients should be derived from visitor-driven revenue rather than per-machine revenue.

Effective Allocatable Space

The total venue space is 3,500 sq ft. The back-of-house allocation of 320 sq ft is fixed and removed from the decision space. A 12% circulation factor is applied to account for aisles, transitions, and non-allocatable space inherent in any venue layout. The effective allocatable space available to the four decision zones is therefore:

$$S_{eff} = (3500 / 1.12) - 320 = 3125 - 320 = 2{,}805 \text{ sq ft}$$

Annual Revenue by Zone

Each zone's annual revenue contribution is derived from the fixed parameters before computing the per-square-foot coefficient.

Entry fee revenue (attributed to FreePlay Zone):

$$R_{entry} = V \times e \times d = 40 \times 12.50 \times 364 = \$182{,}000$$

Premium play revenue (attributed to Premium Zone):

$$R_prem = V \times p \times d = 40 \times 8.00 \times 364 = \$116,480$$

Bar revenue (attributed to Bar Zone):

$$R_bar = V \times b \times d = 40 \times 12.00 \times 364 = \$174,720$$

Bay revenue per bay per year (booking revenue plus bar supplement):

$$\begin{aligned} R_bay &= (r \times w \times u \times 52) + (28 \times w \times u \times 52) \\ &= (95 \times 3.5 \times 0.50 \times 52) + (28 \times 3.5 \times 0.50 \times 52) \\ &= \$8,645 + \$2,548 = \$11,193 \text{ per bay per year} \end{aligned}$$

Revenue Coefficients ($ per sq ft per year)

The objective function coefficients are derived by dividing each zone's annual revenue by its baseline square footage allocation:

Zone	Annual Revenue	Baseline Sq Ft	Coefficient ($/sq ft/yr)
FreePlay Floor (c_1)	$182,000	975 sq ft (15 machines * 65)	$186.67
Premium Room (c_2)	$116,480	390 sq ft (6 machines * 65)	$298.67
Private Bays (c_3)	$11,193/bay	220 sq ft/bay	$50.88
Bar + Entry (c_4)	$174,720	550 sq ft	$317.67

Linear Programming Formulation

Decision Variables:

x_1 = sq ft allocated to FreePlay floor
x_2 = sq ft allocated to Premium room
x_3 = sq ft allocated to Private bays
x_4 = sq ft allocated to Bar + entry + bathrooms

Objective Function:

$$\text{Maximize } Z = 186.67x_1 + 298.67x_2 + 50.88x_3 + 317.67x_4$$

Constraints:

Space constraint — total allocations cannot exceed effective space:

$$x_1 + x_2 + x_3 + x_4 \leq 2{,}805$$

Zone bounds — minimum viable and maximum practical allocations:

$$650 \leq x_1 \leq 975 \quad \text{(10 to 15 FreePlay machines × 65 sq ft)}$$
$$260 \leq x_2 \leq 520 \quad \text{(4 to 8 Premium machines × 65 sq ft)}$$
$$0 \leq x_3 \leq 1{,}760 \quad \text{(0 to 8 bays × 220 sq ft)}$$
$$400 \leq x_4 \leq 700 \quad \text{(functional bar minimum to maximum)}$$

Non-negativity: $x_1, x_2, x_3, x_4 \geq 0$

Solution

MATLAB Implementation:

The LP was solved using MATLAB's linprog function. Since linprog minimizes by default, the objective coefficients were negated to convert the maximization problem:

```
[x_opt, fval] = linprog(-c, A, b, [], [], lb, ub, options)
```

where c = [186.67; 298.67; 50.88; 317.67], A = [1 1 1 1], b = 2805, and lb/ub are the bound vectors. Optimal revenue is recovered as Z = -fval. The complete implementation including graphical output is included as the accompanying .m file.

Optimal Solution:

Zone	Optimal Allocation	Baseline Sq Ft	Coefficient ($/sq ft/yr)
FreePlay Floor (x_1)	975 sq ft	15 machines	$182,000
Premium Room (x_2)	520 sq ft	8 machines	$155,307
Private Bays (x_3)	610 sq ft	3 bays	$31,035
Bar + Entry (x_4)	700 sq ft	—	$222,371
Back of house	320 sq ft (fixed)	—	—
TOTAL	3,125 sq ft	23 machines + 3 bays	$590,713

The LP allocated FreePlay, Premium, and Bar to their maximum bound. Bar ($317.67/sq ft) and Premium ($298.67/sq ft) have the highest coefficients and reach their upper bounds first. FreePlay ($187.67/sq ft) also reaches its upper bound as entry fee revenue is attributed entirely to this zone. Bays ($50.88/sq ft) receive only the space that remains after higher-yield zones are satisfied, settingly at 3 bays rather than the theoretical maximum of 8.

Feasible Region

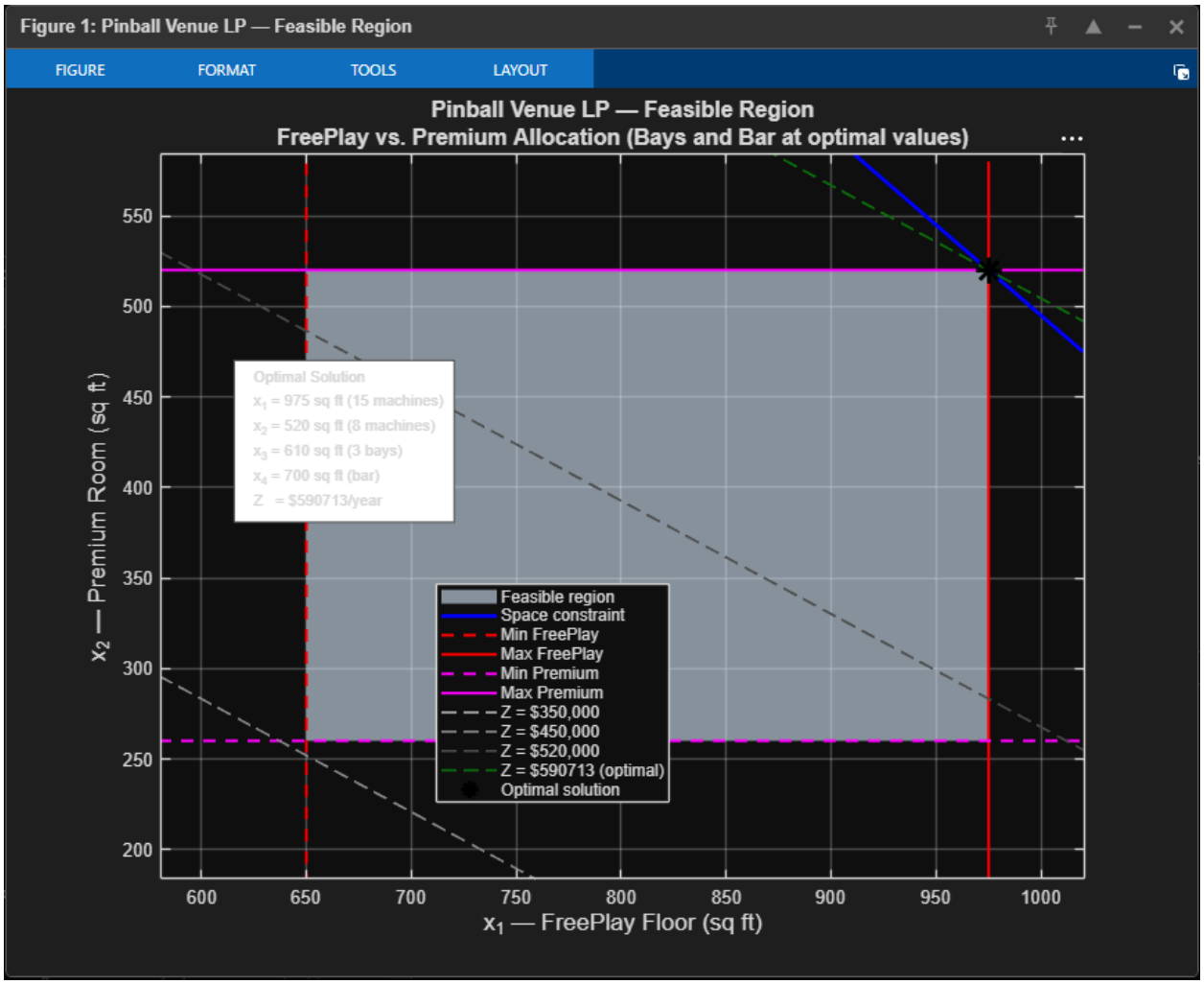

Figure 1 (generated by the MATLAB script) shows the two-dimensional feasible region for x_1 vs. x_2 with bays and bar fixed at their optimal values. Iso-revenue contour lines show the objective function increasing toward the optimal corner point at (975, 520), where the Max FreePlay, Max Premium, and space constraints are simultaneously binding.

Revenue Hierarchy at Optimal Solution

Revenue Stream	Annual Amount	% of Total
Bar/beverage (x_4)	$222,371	37.6%
Entry fees – FreePlay (x_1)	$182,000	30.8%
Premium play (x_2)	$155,307	26.3%
Private bays (x_3)	$31,035	5.3%

TOTAL	$590,713	100%

Bar and entry fees together account for 68.4% of total revenue. Bays contribute only 5.3% despite requiring 610 square feet of dedicated build-out space, confirming the conclusion from the pre-Linear Programming simulation model that bays have a structural revenue ceiling that visitor-driven zones do not.

Limitations and Conclusion

Limitations

The most significant limitation is the treatment of bay revenue as linear in square footage. In practice, bays have a hard revenue ceiling determined by bookable hours and demand, a non-linearity the pre-LP simulation model explicitly demonstrated. A more rigorous solution would require piecewise linear approximation or non-linear programming. Visitor count is also treated as a fixed parameter rather than a function of venue configuration, which understates the feedback between floor quality and attendance. All coefficients are derived from modeled estimates rather than empirical operational data and should be validated against real location-specific inputs before use in a formal financial projection.

Conclusion

This project applied Linear Programming to determine the optimal floor space allocation for a proposed 3,500 square foot pinball venue. The LP formulation produced a well-defined objective function maximizing annual gross revenue across four decision variables, subject to a binding space constraint and eight bound constraints reflecting minimum viable and maximum practical zone sizes.

The optimal solution allocates FreePlay, Premium, and Bar zones to their maximum bounds, with Private Bays receiving the remaining 610 square feet. Optimal annual revenue at 40 average daily visitors is $590,713, with the bar area contributing 37.6%, entry fees 30.8%, premium play 26.3%, and bays 5.3%. The result formally confirms what the earlier simulation and business analysis had suggested qualitatively: bar and floor revenue dominate, and over-allocating space to bays at the expense of higher-yield zones is suboptimal.

The convergence of three independent analytical approaches, interactive simulation, business reasoning, and formal LP optimization, on the same structural conclusion is the strongest validation that this model can offer at a pre-operation planning stage. Ultimately, the LP produced a result that had only been theorized about through the previous simulation, but was now formally verified through the optimization model. For me personally, this has provided real eye-opening insight into the practical applications of Linear Programming through the intersection of academia and my personal hobbies/interests and for that I am very grateful.

MATLAB Output:

```
Revenue coefficients ($/sq ft/year):
FreePlay (c1): $186.67
Premium (c2): $298.67
Bays (c3): $50.88
Bar (c4): $317.67
============================================
OPTIMAL SOLUTION
============================================
FreePlay floor : 975.0 sq ft (15 machines)
Premium room : 520.0 sq ft (8 machines)
Private bays : 610.0 sq ft (3 bays)
Bar + entry : 700.0 sq ft
Back of house : 320.0 sq ft (fixed)
--------------------------------------------
Total allocated: 3125.0 sq ft
--------------------------------------------
Optimal Annual Revenue: $590713
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Revenue breakdown:
Entry fees (FreePlay) : $182000 (30.8%)
Premium play : $155307 (26.3%)
Private bays : $31035 (5.3%)
Bar / beverage : $222371 (37.6%)
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Total : $590713
Figure generated: Feasible region — FreePlay vs. Premium allocation
```